

## Summary

Data Scientist with 1 year of experience building machine learning models, data pipelines, and AI-powered applications in Python. My work spans forecasting, analytics, LLM-based systems, and backend development, with a focus on turning data and models into practical software. I enjoy working across the full lifecycle, from exploring and modelling data to building APIs, integrating AI workflows, and deploying working systems.

## Professional Experience

### Data Scientist

*Arabania Indo-Arabic Restaurant*

*June 2024 - May 2025, Lucknow*

- Built demand forecasting models in Python using statsmodels and scikit-learn across 100K+ inventory records, helping reduce food waste by 20% and procurement costs by 12% through more accurate daily inventory planning.
- Ran end-to-end A/B pricing experiments using hypothesis testing and statistical inference, identifying a pricing strategy that improved gross margins by 15% within 2 months, while also building SKU performance models that improved operational efficiency by 12%.
- Built automated SQL analytics pipelines using CTEs and window functions across 100K+ records to identify demand and seasonality patterns, reducing manual reporting time by 40% through full automation.

## Projects

### AI Competitor Research Platform

Live Demo   GitHub

*Python, FastAPI, React, Firecrawl, Playwright, PostgreSQL, SQLite, RAG, LLM APIs, Docker*

- Built an evidence-backed competitive intelligence platform that orchestrates URL discovery, region-aware research, web scraping, structured LLM extraction, validation, persistent storage, cross-run analysis, reporting, and alerts.
- Implemented a layered scraping architecture using Firecrawl, Playwright, and HTTP fallback, with multi-provider LLM support across Gemini, OpenAI, Anthropic, and Groq plus provider fallback and cooldown handling.
- Built schema and semantic validation with corrective extraction, evidence retention, and run-scoped observations, preserving source URLs, timestamps, regions, scraped content, and content hashes for traceable competitive findings.
- Developed the FastAPI backend and React intelligence workspace for research execution, evidence exploration, competitive signals, review-based RAG, and report generation, with deterministic demo support for credential-free evaluation.

### RouteForge – Intelligent Model Routing System

Live Demo   GitHub

*Python, FastAPI, LightGBM, NumPy, Pydantic, REST APIs, JavaScript, Chrome Extension*

- Built a provider-independent model routing backend around the core S2 Sweep implementation, exposing model-selection decisions through a FastAPI service without coupling the router to LLM inference providers.
- Implemented controlled routing exploration using a 70% quality-cost selection path and 30% uniform exploration across eligible models, returning the selected model, scores, and routing metadata through the API.
- Developed a command-line interface, browser console, and Chrome/Chromium Manifest V3 integration for ChatGPT and Claude, with an explicit compatibility layer separating historical model IDs from provider-specific model labels.
- Added conservative browser handoff and optional auto-submit controls, with safe fallback behavior when provider mappings or visible model controls cannot be identified confidently.

### TrustLayer - AI Agent Authorization API

GitHub

*FastAPI, PostgreSQL, SQLAlchemy, Alembic, React, Docker Compose, Pytest, Testcontainers*

- Built an authorization backend for AI-agent workflows using first-match policy evaluation with default-deny behavior and explicit ALLOW, DENY, and REQUIRE\_APPROVAL decisions.
- Implemented organization-scoped policies, agent-scoped machine credentials, credential rotation and revocation, approval queues, approval expiration, and explicit approve/reject workflows through the API and React dashboard.
- Designed append-only audit logging with database-level protection against audit-record updates and deletes, preserving authorization and approval history for traceability.
- Added backend unit and PostgreSQL integration tests with Testcontainers, frontend tests, type checking, production builds, Docker Compose infrastructure, and demo seed flows for end-to-end verification.

## Technical Skills

- **AI/ML:** scikit-learn, XGBoost, LightGBM, Logistic Regression, time-series forecasting, feature engineering, model validation, cross-validation, anomaly detection, backtesting, statistical inference
- **LLMs and Agents:** OpenAI, Anthropic, Gemini, Groq, LangChain, RAG, structured LLM extraction, custom agent development, prompt engineering, provider routing, corrective extraction, structured output validation
- **Backend and APIs:** Python, FastAPI, Flask, REST APIs, Pydantic, SQLAlchemy, PostgreSQL, SQLite, Alembic, Docker, Docker Compose, API authentication, caching
- **Data and Analytics:** Pandas, NumPy, statsmodels, SQL, CTEs, window functions, query optimisation, automated data pipelines, Power BI, Tableau, Matplotlib, Seaborn, Jupyter
- **Frontend and Engineering:** React, JavaScript, Streamlit, Git, GitHub Actions, pytest, Testcontainers, Chrome Extension development

## **Certifications**

- Google Data Analytics Professional Certificate (Foundations, Data Exploration, Data-Driven Decisions)

## **Education**

**Indian Institute of Technology, Madras**

*2022-2025*

B.S. in Data Science and Applications